ENTERPRISE AI INFRASTRUCTURE • OPEN SOURCE PLATFORM

BIFROST GATEWAY

One Gateway. Every AI. Infinite Possibilities.

Unified Multi-Provider API • Intelligent Model Routing • pgvector Semantic Cache • 18ms Failover • MCP Tool Gateway

< 4ms	71%	15+	99.99%
Internal Latency	Cost Reduction	Providers Connected	Uptime SLA

MARKET OPPORTUNITY

The Enterprise AI Infrastructure Crisis

Why direct multi-provider API integration creates operational chaos, outages, and runaway costs.

CRITICAL FRICTION

Vendor Lock-in & Sprawl

Teams separately integrate OpenAI, Anthropic, Gemini, DeepSeek, and Bedrock.

Inconsistent SDKs, divergent error codes, and unique streaming transcoders.

Engineering time wasted rewriting client code whenever new models release.

Zero single-pane-of-glass visibility across enterprise LLM traffic.

SLA BREACHES

Unreliable Upstream Outages

Rate limits (HTTP 429) and provider outages (503/500) crash production apps.

Developers build brittle in-app retries that cause destructive retry storms.

Lack of dynamic failover between cloud regions and alternative providers.

No circuit breaker protection against runaway looping agents.

BUDGET WASTE

Runaway Token Spend

Identical and repetitive queries repeatedly billed at full frontier prices.

Simple classification tasks routed to expensive flagship models.

Absence of semantic caching and per-tenant budget kill-switches.

API keys hardcoded across microservices with zero access control.

PRODUCT VISION

The Bifrost Gateway Solution

One unified high-throughput infrastructure platform connecting applications with all AI providers.

STANDARDIZED CONTRACT

Unified Drop-in OpenAI API

Developers point any OpenAI client to Bifrost by updating base_url.

Zero code rewrite required across Python, TypeScript, Java, and Go.

Standardized REST, SSE streaming, and multimodal payload encoding.

SUB-3MS DECISION

Intelligent Dynamic Router

Evaluates prompt intent, context window, latency SLAs, and cost ceilings.

Automatically routes each request to the optimal model in < 3ms.

Weighted load balancing and tag-based policy distribution.

ZERO-COST INFERENCE

pgvector Semantic Caching

Hybrid exact match and cosine similarity caching using PostgreSQL & Redis.

Serves repeated or semantically equivalent queries in < 4ms at $0 cost.

Slashes enterprise monthly token bills by 35% to 72%.

100% 200 OK DELIVERY

18ms Disaster Recovery Failover

Integrated circuit breakers detect 429 rate limits or upstream 503 outages.

Instantly falls back to secondary frontier models with 0 app downtime.

Enterprise Virtual Keys with monthly spend caps and PII redaction.

ENGINEERING DESIGN

Complete 6-Layer System Architecture

High-throughput Go data plane separated from the administrative control plane.

LAYER 01 Client & Ingress Layer

Web/Mobile Apps, Python/TS SDKs, Autonomous Agents, Cursor, LangChain

LAYER 02 Gateway Edge & Security

Go FastHTTP reverse proxy, Token-bucket rate limiting, PII/PCI redaction

LAYER 03 Intelligence & Routing Engine

pgvector semantic distance lookup, Multi-objective router, Circuit breakers

LAYER 04 Provider Adapter Matrix

OpenAI, Anthropic, Gemini, DeepSeek, Groq, Bedrock, Azure, Mistral, Ollama

LAYER 05 Distributed Data & Cache

PostgreSQL 16 pgvector, Redis 7 cluster for rate limits, AES-256 key vault

LAYER 06 Operations & Governance

OpenTelemetry distributed W3C tracing, Prometheus /metrics, 14-screen Console

ECOSYSTEM COMPATIBILITY

Universal Multi-Provider Coverage

Seamlessly interconnect 15+ commercial frontier, open-weight, and on-premise inference engines.

OpenAI

GPT-4o, GPT-4o-mini, o1, o3-mini

Latency: 142ms • Pricing: $2.50 / $10.00

Anthropic Claude

Claude 3.5 Sonnet, 3.5 Haiku, Opus

Latency: 168ms • Pricing: $3.00 / $15.00

Google Gemini

Gemini 1.5 Pro (2M Context), 2.0 Flash

Latency: 118ms • Pricing: $1.25 / $5.00

DeepSeek

DeepSeek R1 (Reasoning), DeepSeek V3

Latency: 95ms • Pricing: $0.14 / $0.28

Groq LPU

Llama 3.3 70B (480 tokens/sec)

Latency: 38ms • Pricing: $0.59 / $0.79

AWS Bedrock

Claude 3.5, Titan, Command R+

Latency: 185ms • Pricing: Enterprise IAM

Azure OpenAI

GPT-4o Enterprise Private Endpoint

Latency: 154ms • Pricing: HIPAA BAA

Mistral AI

Mistral Large 2, Codestral 2501

Latency: 122ms • Pricing: EU Sovereign

Ollama Local

Llama 3.2, Qwen 2.5 (Air-Gapped)

Latency: 62ms • Pricing: $0.00 (Self-Hosted)

TRAFFIC OPTIMIZATION

Intelligent Model Routing Engine

Dynamic, multi-objective optimization routes requests to the ideal model in under 3 milliseconds.

DYNAMIC MULTI-OBJECTIVE

Routing Policies & Strategies

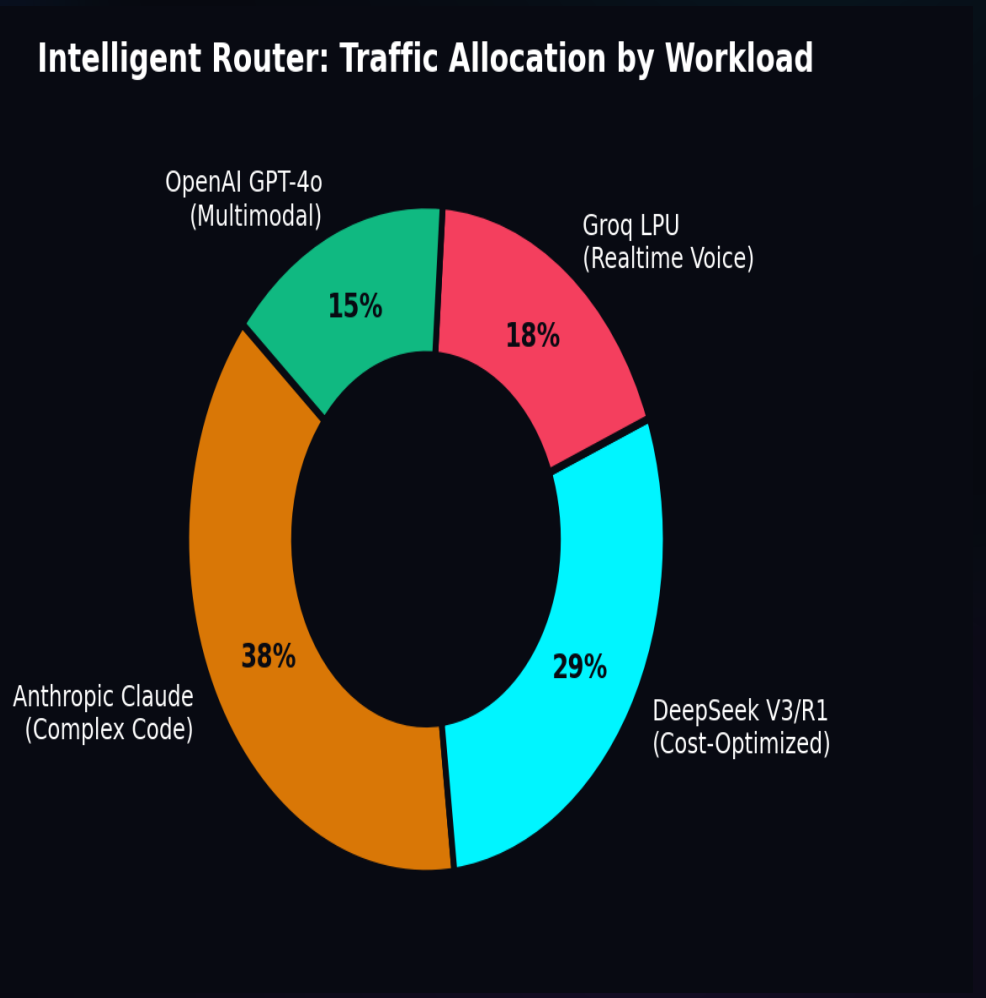
Cost-Optimized Strategy: Enforces max $0.50/1M constraint, routing routine queries to DeepSeek V3 or Gemini Flash (-78% spend).

Sub-50ms Latency Strategy: Routes conversational voice workloads to Groq LPU (38ms P99) for immediate speech output.

High-Context Coding Strategy: Automatically sends complex AST synthesis and refactoring to Claude 3.5 Sonnet.

Long-Form Document Strategy: Utilizes Gemini 1.5 Pro's native 2M token context window without brittle chunking.

Weighted Enterprise Balance: Splits volume across Azure OpenAI (50%), Bedrock (30%), and Google Cloud (20%).



HIGH AVAILABILITY

Zero-Downtime Automatic Failover (18ms)

Circuit breakers and bounded fallback chains eliminate 99.9% of upstream AI downtime.

CIRCUIT BREAKER PROTECTION

Disaster Recovery Invariants

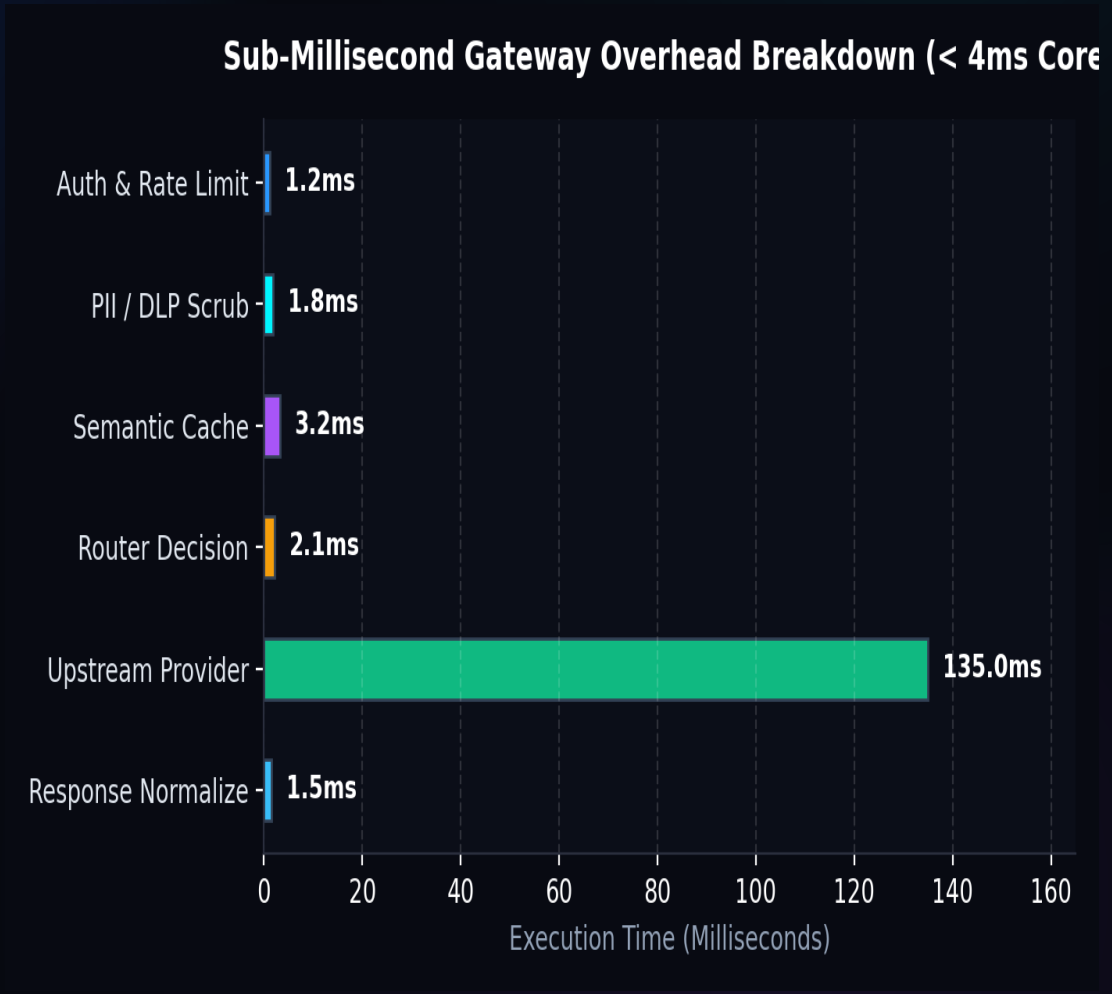
Upstream Outage Interception: Catches HTTP 429 (Rate Limits) and 503/500 errors before applications notice.

18ms Instant Fallback: Cascades to alternative provider (e.g. OpenAI -> Claude 3.5 -> Azure OpenAI) in under 20ms.

Circuit Breaker State Machine: Trips to OPEN state after 3 consecutive failures with 30s cooldown timer.

Bounded Retry Budget: Clamps retries to prevent catastrophic downstream retry storms.

100% Client Availability: Returns HTTP 200 OK without surfacing provider downtime to users.



ECONOMIC EFFICIENCY

Cost Intelligence & Semantic Vector Caching

Cut enterprise AI API expenditures by 71% through intelligent memory and model tiered routing.

VECTOR MEMORY LAYER

Semantic Cache Architecture

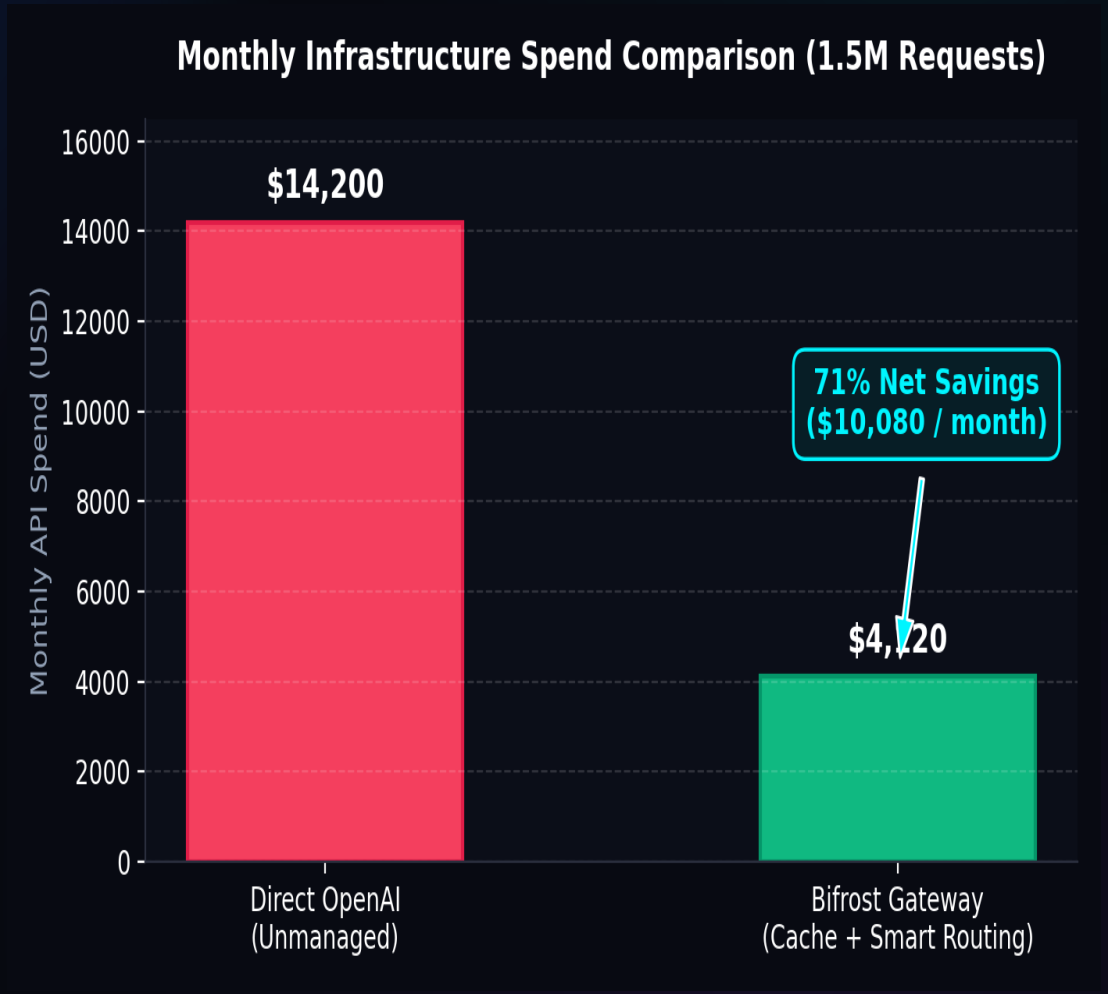
Cosine Distance Lookup: Embeds queries and queries pgvector IVFFlat index in under 4ms.

Configurable Similarity Threshold: Dial between 0.80 (loose high-hit) to 0.99 (strict near-exact) precision.

Zero-Cost Responses: 35% of repetitive queries answered with 0 upstream tokens billed.

TTL & Invalidation: Automatic time-to-live policies and tenant-isolated namespace boundaries.

Annualized ROI: Saves $120,000+ per year on high-volume 1.5M requests/month workloads.



PERFORMANCE BENCHMARKS

Latency Spectrum & Real-Time Performance

Sub-4ms internal gateway overhead with hardware-accelerated LPU streaming.

SUB-4MS ARCHITECTURE

Benchmark Characteristics

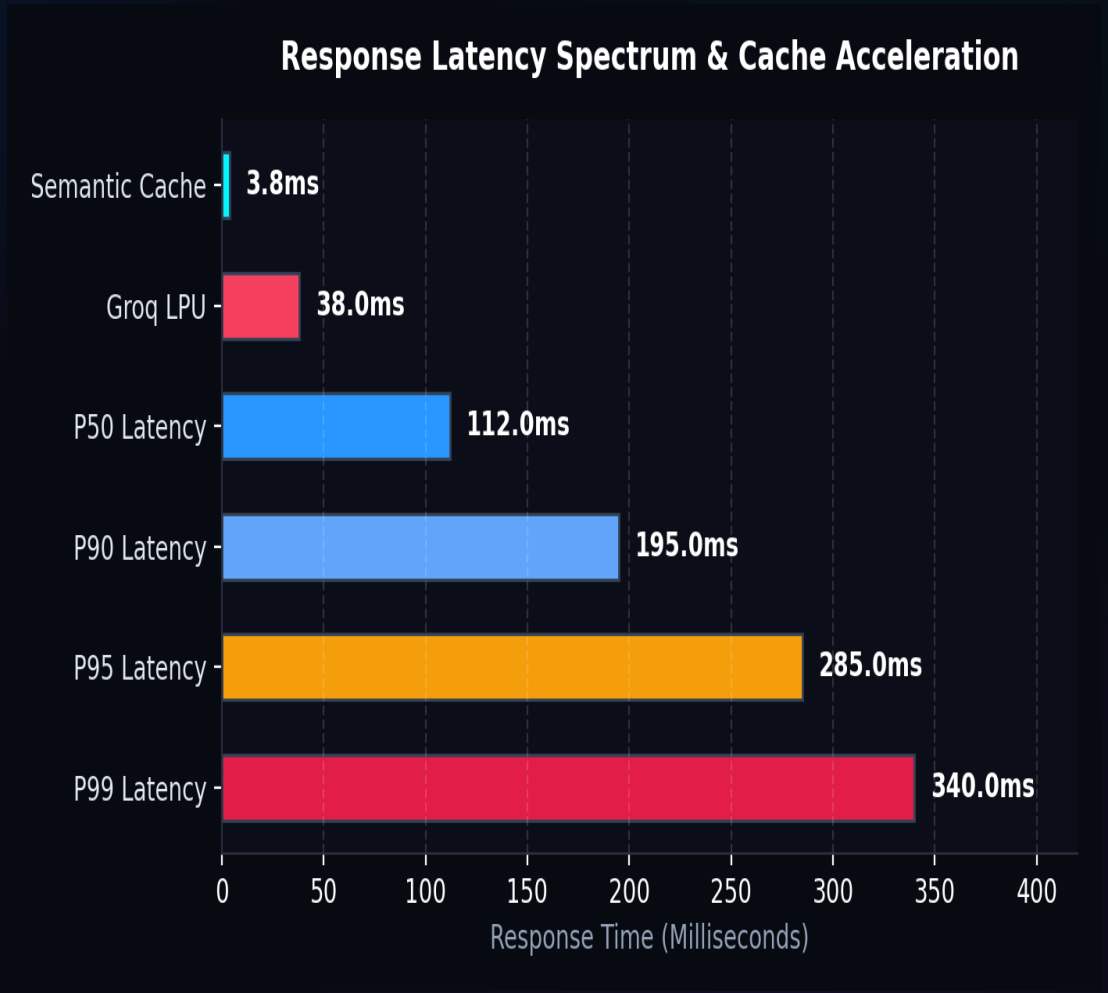
Internal Overhead: < 4ms added latency for full authentication, caching, guardrails, and routing.

Semantic Cache Speed: 3.8ms total roundtrip response for vector cache hits.

Groq LPU Integration: 38ms time-to-first-token with 480 tokens/sec streaming throughput.

P95 / P99 SLA: P50 latency 112ms, P99 guaranteed under 340ms across multi-cloud regions.

SSE Streaming Transcoder: Converts divergent upstream streams to standardized SSE in < 1.2ms.



DATA PROTECTION

Enterprise Security, Virtual Keys & Guardrails

Never expose root provider secrets. Issue scoped virtual keys with automated budget kill-switches.

TOKEN SCOPING

Tenant Virtual API Keys

Issue isolated virtual keys to specific teams, microservices, or frontend clients.

Clamp monthly budgets (e.g. $2,500/mo) with automated reject upon ceiling.

Restrict keys to specific allowed model whitelists and RPM rate limits.

PRIVACY SHIELD

Real-time DLP & PII Redaction

Sanitize credit card PANs, SSNs, and patient identifiers before packets leave VPC.

Full HIPAA, GDPR, and PCI-DSS compliance conformance ready.

Custom regex and embedding-based sensitive entity maskers.

THREAT MITIGATION

Prompt Injection Defense

Block adversarial jailbreak attempts ('Ignore all previous instructions...').

Detect toxic outputs and hate speech with real-time classification gates.

Cost velocity spike breaker ($50/min cap) prevents infinite agent loops.

AUDIT COMPLIANCE

SOC2 Immutable Audit Trails

Every request, key rotation, policy change, and PII redaction event logged.

Full W3C trace context propagation across every microservice hop.

Export audit records to Splunk, Datadog, or AWS S3 cold storage.

AUTONOMOUS AGENTS

Model Context Protocol (MCP) Tool Bridge

Governed tool execution and human-in-the-loop approval gates for agentic workflows.

AGENT CLIENTS

Agentic Integrations

Direct bridge for LangChain, CrewAI, AutoGen, Cursor, and Claude Code.

Agents connect to Bifrost via standardized OpenAI tool-calling semantics.

Eliminates unmonitored raw shell or direct production database access.

Detailed tool call latency, token usage, and error metrics recorded.

ENTERPRISE TOOLS

Registered MCP Tools

PostgreSQL Enterprise MCP: Read-only parameterized queries with limit clamps.

GitHub Org MCP: Secure PR creation and issue tracking with branch protection.

Slack Alerts MCP: Incident escalation dispatch to #ai-ops channels.

Brave Web Search MCP: Real-time web index search with content extraction.

HUMAN GATES

Approval Gates & Governance

Granular Execution Policies: Toggle tools between 'Auto-Approved' vs 'Approval Required'.

Human-in-the-Loop: High-risk tools (e.g. create PR, send alert) require explicit admin sign-off.

Parameter Validation: JSON schema enforcement blocks SQL injection attempts.

Complete Audit Playback: Replay agent execution steps for incident forensics.

PRODUCTION READY

Deployment, Administration & Open-Source Roadmap

Enterprise self-hosting with Docker, Kubernetes, and 14-screen admin console.

TURNKEY DEPLOYMENT

Self-Hosted Deployment Stack

Docker Compose: Single command bootstrap with Redis and pgvector.

Kubernetes & Helm: Multi-replica deployment, mTLS ingress, and HPA auto-scaling.

22MB Static Go Binary: Zero external runtime dependencies for maximum security.

CONTROL PLANE

14-Screen Admin Console

Built-in Next.js 15 dark-mode control plane at /dashboard.

Live waterfall trace inspector with sub-millisecond execution spans.

Configurable semantic similarity thresholds and one-click cache purges.

EXECUTIVE ROI

Proven Economic ROI

71% reduction in monthly API spend ($10,080/mo saved on 1.5M requests).

Zero vendor lock-in allows negotiating volume discounts across providers.

SLA protection prevents catastrophic customer churn during provider outages.

GET STARTED TODAY

Open Source & Next Steps

Released under Apache 2.0 open-source license on GitHub.

Explore live interactive playground at http://localhost:3000/playground.

GitHub Repository: https://github.com/Ranjeet7680/BIFROST-GATEWAY.